Programmable Matrix Mathematics for Industrial Sensor Fusion

Complete SPD matrix operations, digital-twin covariance fusion and versioned mathematical quality rules



Concept illustration; not a computed research result.

Verified outcome

Using the original paceval C++ SDK 4.25, new complete 128 × 128 and 512 × 512 dense symmetric positive-definite matrices were factored and inverted. A synthetic, six-dimensional industrial sensor-fusion model was executed in 32 versioned scenarios. The 512-dimensional run used 17 persistent graphs and produced 67,538,758 actual paceval graph outputs.

Independent full-matrix and sensor-fusion checks have passed. The historical 1024-dimensional experiment described in the supplied source report is background and is not counted as newly executed.

13 November 2026 | Functionality and feasibility | Synthetic data | No performance comparisons

1. Introduction: Industrial use case

A manufacturing inspection cell may receive overlapping observations from laser scanners, thermal sensors, coordinate measurement systems and machine-state estimators. Each measurement has an uncertainty model. A digital twin that combines such information needs covariance matrices, reliable matrix factorizations and a consistently applied mathematical definition of acceptable disagreement.

The present research illustrates a supervisory precision-metrology application: two six-dimensional synthetic sensors observe the same nominal state; their covariance models encode uncertainty and correlation. A mathematical fusion algorithm estimates a common state, then computes a normalized innovation statistic, uncertainty radius, versioned quality scores and signed margins. The output is a repeatable quality report, not direct control of an actuator.

Why a mathematical execution engine is relevant

The base matrix operations are well-established. The research question is whether those operations, plus frequently modified application-specific equations and score rules, can be represented as reusable mathematical execution graphs. The algorithm preserves the distinction between the general matrix decomposition, the application-layer uncertainty equations, and a final host-side quality decision.

Current scientific and industrial context

LAPACK provides standard Cholesky decomposition and SPD solving [1,2]. Recent research on square-root visual-inertial estimation (2025) illustrates the importance of structured positive-definite covariance representations [3]. Multisensor data fusion in CNC digital twins has been reviewed in 2025 [4]. NIST research, including a 2026 workshop report, emphasizes uncertainty quantification, digital-twin credibility and verification/validation throughout the model lifecycle [5,6].

The present study does not claim to invent sensor fusion or matrix decomposition. It establishes which mathematical operations were actually performed using paceval, and how their outputs can be reused in versioned, application-specific formulas.

2. Research question and execution architecture

Core question. Can the original paceval library actually compute a full general dense SPD Cholesky factorization and inverse, and then execute complete sensor-fusion functions and versioned quality rules by reusing mathematical graphs?

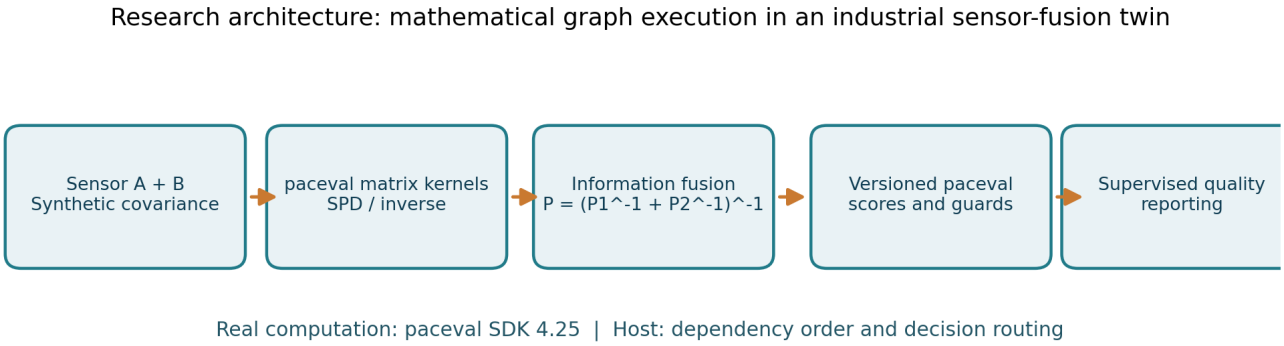


Figure 1. Research architecture. Data are synthetic; paceval executes mathematical operations, while C++ controls operation order and final policy.

Layer	Actual responsibility
Input data	Generate deterministic SPD matrices, synthetic sensor observations and documented model versions.
Original paceval SDK 4.25	Evaluate square-root, division, Schur updates, inverse assembly, covariance fusion, mathematical scores and sensitivity graphs.
C++ orchestrator	Order dependent pivots, manage matrices and batches, validate domains and select decision labels.
Independent scientific audit	Verify every complete matrix entry with NumPy/LAPACK and inspect 80-decimal-digit samples using mpmath.

The research software is a real original SDK integration. No imitation paceval engine, locally precomputed result substitution or hidden external solver is used in the productive C++ matrix path.

3. Complete mathematical definitions

Full symmetric positive-definite decomposition

For a real SPD matrix A, a lower-triangular factor L exists with positive diagonal entries such that $A = L\,L^{T}$. A right-looking update forms the diagonal pivot, computes the column below the pivot and updates the trailing Schur complement. These operations run through persistent paceval graphs: sqrt(a), a/b and a-b*c.

After Cholesky decomposition, the program constructs $T = L^{-1}$ by forward elimination. It then assembles the complete inverse $A^{-1} = T^{T}T$. The multiplication and accumulation are evaluated through the persistent a+b*c graph. Scalar log-determinant terms use ln(a) and a+2*b.

Two-sensor information fusion

For independently modeled sensor estimates m_1 and m_2 with covariance matrices P_1 and P_2, the information-form model is $P_f = (P_1^{-1} + P_2^{-1})^{-1}$, and the fused estimate is $\hat{x} = P_f\,(P_1^{-1}m_1 + P_2^{-1}m_2)$. Covariance formation, matrix/vector products and accumulation use paceval graph calls.

Sensor disagreement is quantified using $d^2 = (m_1 - m_2)^{T}\,(P_1 + P_2)^{-1}\,(m_1 - m_2)$. This covariance describes the difference between independent observations, and is not the fused covariance. The demo treats dimensions and uncertainty scales as normalized synthetic quantities.

Versioned mathematical gates

A version-specific score combines d^2, a quadratic temperature term, an uncertainty-radius term and the sensor-disagreement norm. Two distinct score graphs, two signed-margin graphs and two analytical temperature-derivative graphs coexist. The illustrative gate threshold is 12.592, with a separate host classification into ACCEPT, REVIEW and REJECT. This policy is not a statistically calibrated industrial certification rule.

4. Actual paceval execution and graph responsibilities

The public source files paceval_main.h and paceval_main.cpp are integrated into a CMake-based Linux C++ program. The compatible Linux paceval shared library is provided externally. The SDK original copyright notices are maintained; the relative header include is adjusted for standalone build layout.

Graph family	Purpose
sqrt(a), a/b	Pivot and triangular division
a-b*c	Cholesky Schur and forward-inverse updates
a+b*c	Inverse assembly and matrix/vector accumulation
ln(a), a+2*b	Complete log-determinant
a+b, a-b, a*a, abs(a)	Covariance, innovation and derived sensor measures
Six distinct versioned graphs	Quality scores, signed margins, temperature derivatives

A total of 17 distinct graphs were successfully registered and each one executed for actual model inputs. Matrix dependency scheduling remains in C++ because later pivots depend on earlier updates; C++ does not replace the arithmetic graph kernels with native mathematics.

For matrices of dimension n, complete factors, inverses and input matrices are archived as little-endian IEEE-754 binary64 files. The graph execution inventory and registry record the actual function strings, variable declarations, API call counts and graph-output counts.

5. Verified new large-matrix outcomes

New execution property	128 × 128	512 × 512
Complete inverse entries	16,384	262,144
Distinct persistent paceval graphs	17	17
Actual graph outputs	1,108,678	67,538,758
Full Cholesky reconstruction	Verified	Verified
Full inverse identity	Verified	Verified
Independent NumPy inverse	All entries checked	All entries checked
Sensor fusion scenarios	32	32
Executed quality-surface samples	625	625

Independent numerical verification

An independent implementation loaded every matrix entry and checked the full input SPD matrix, complete Cholesky factor, complete inverse, symmetry, log-determinant and matrix identities against NumPy/LAPACK. An additional independent implementation verified every complete sensor-fusion state, covariance diagonal, model score, signed quality margin and decision. An 80-decimal-place mpmath audit checked 56 values from selected scenarios.

Detailed numerical differences are preserved in the machine-readable audit rather than displayed as C++-versus-paceval accuracy comparisons in this public whitepaper. Results refer to well-conditioned synthetic matrices; general ill-conditioned or indefinite matrices are not certified by these tests.

The supplied source report additionally describes a historical complete 1024-dimensional experiment. That earlier result is not presented as a new run of this project.

6. Dense matrix input and complete factor

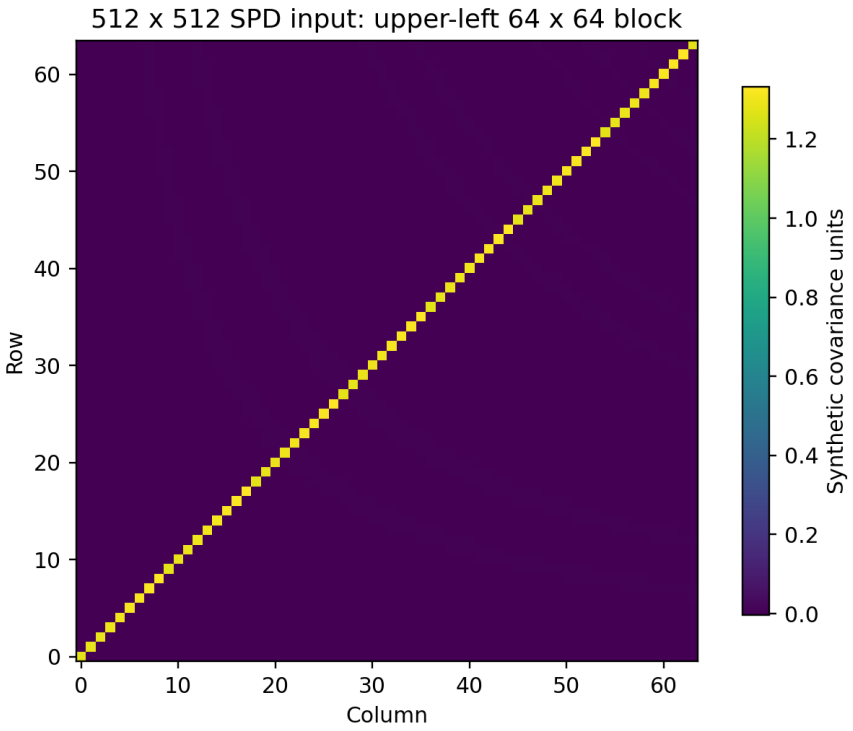


Figure 2. Upper-left block of the complete synthetic 512-dimensional SPD input. The full 512 × 512 matrix is available as binary64 data.

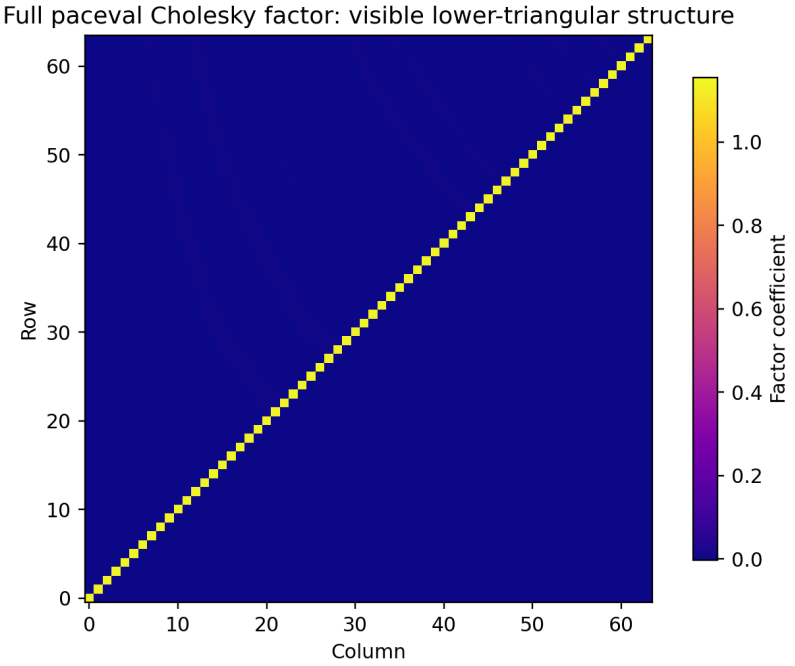


Figure 3. Upper-left block of the complete paceval-calculated lower-triangular Cholesky factor.

7. Complete inverse and covariance interpretation

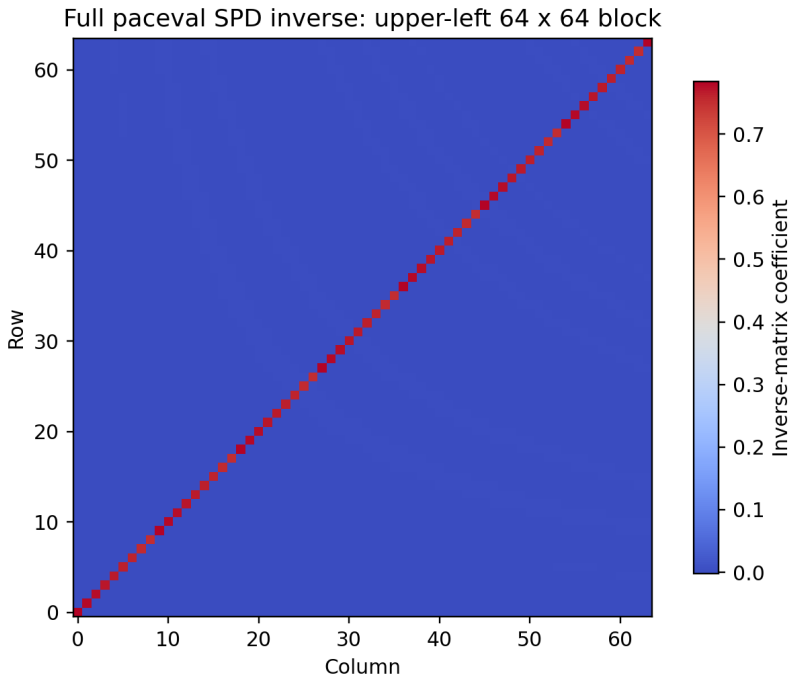


Figure 4. Upper-left block of the complete paceval-calculated SPD inverse, not a selected-entry calculation.

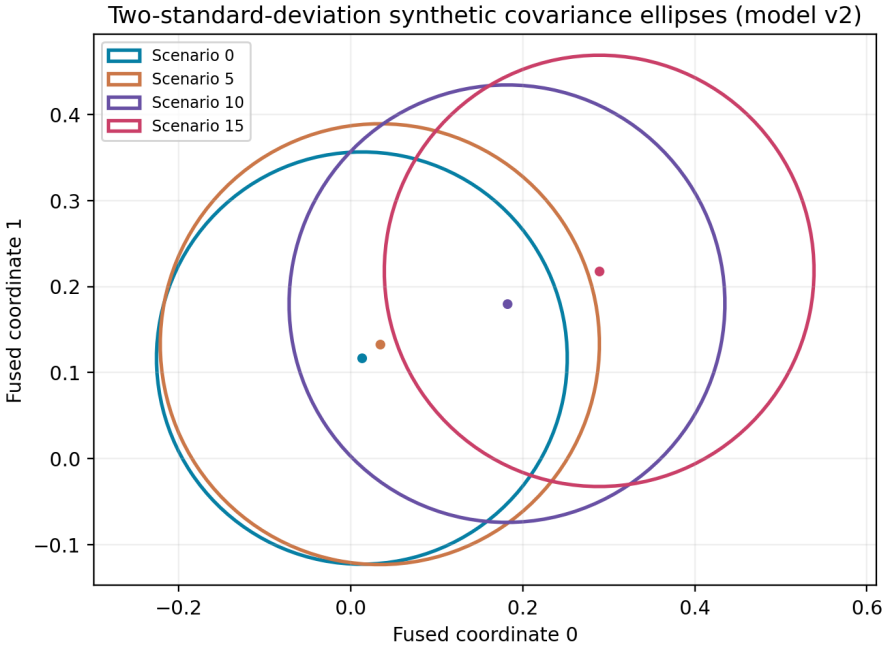


Figure 5. Illustrative two-coordinate covariance ellipses from selected full six-dimensional paceval sensor-fusion results.

8. Fusion results and innovation gating

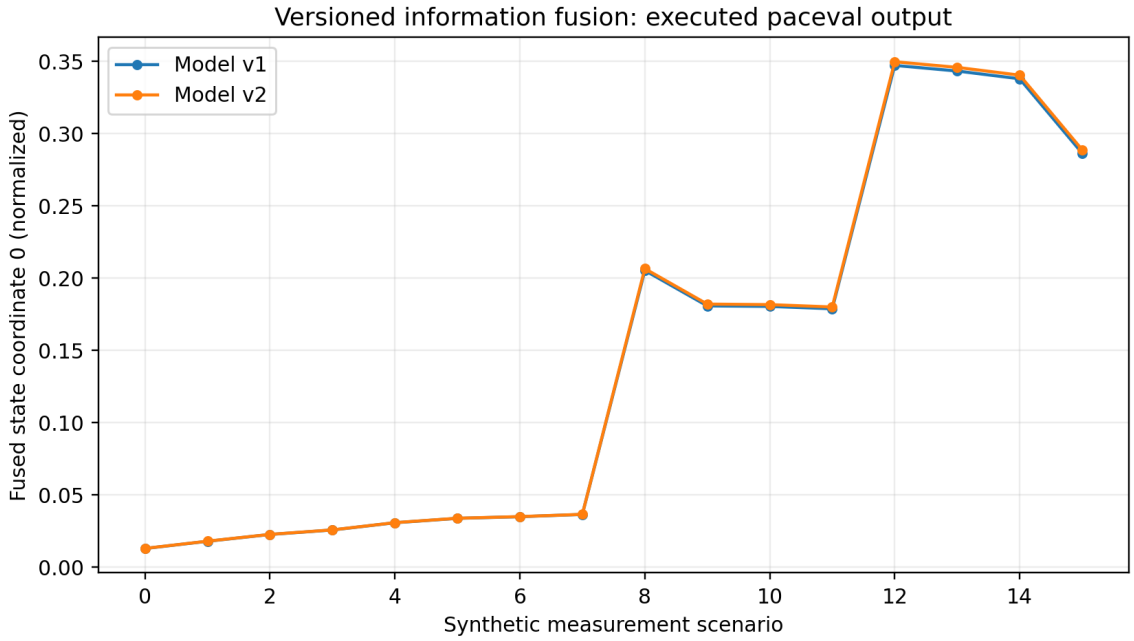


Figure 6. Fused first state coordinate from 16 synthetic measurement scenarios across both preserved model versions.

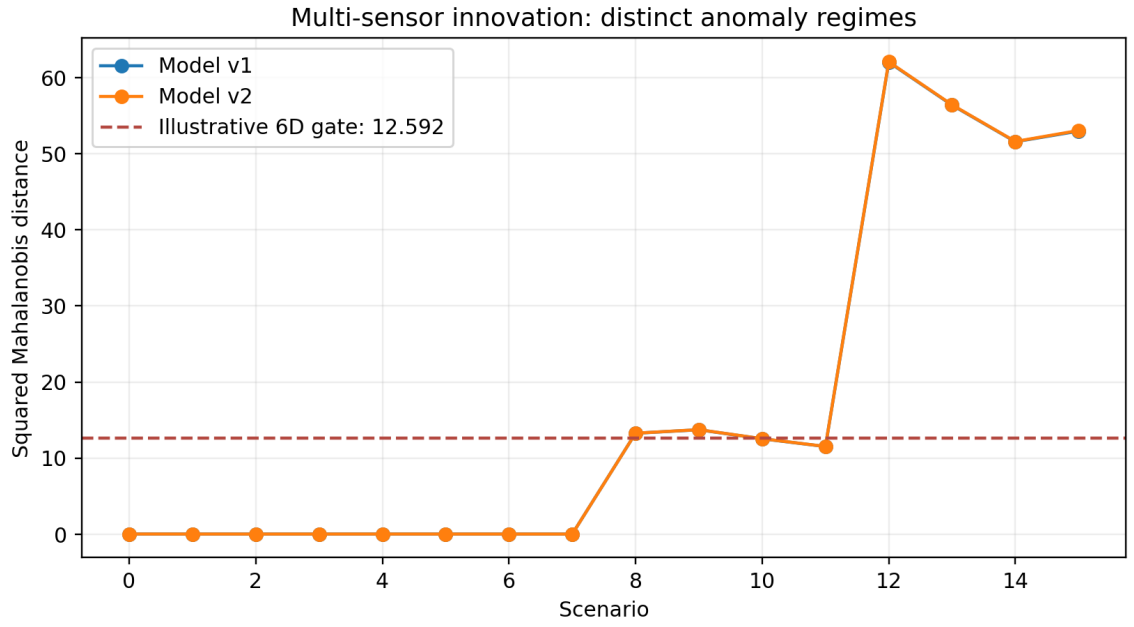


Figure 7. Normalized sensor innovation across synthetic operating scenarios; large disagreement leads to flagged cases.

9. Versioned mathematical quality decisions

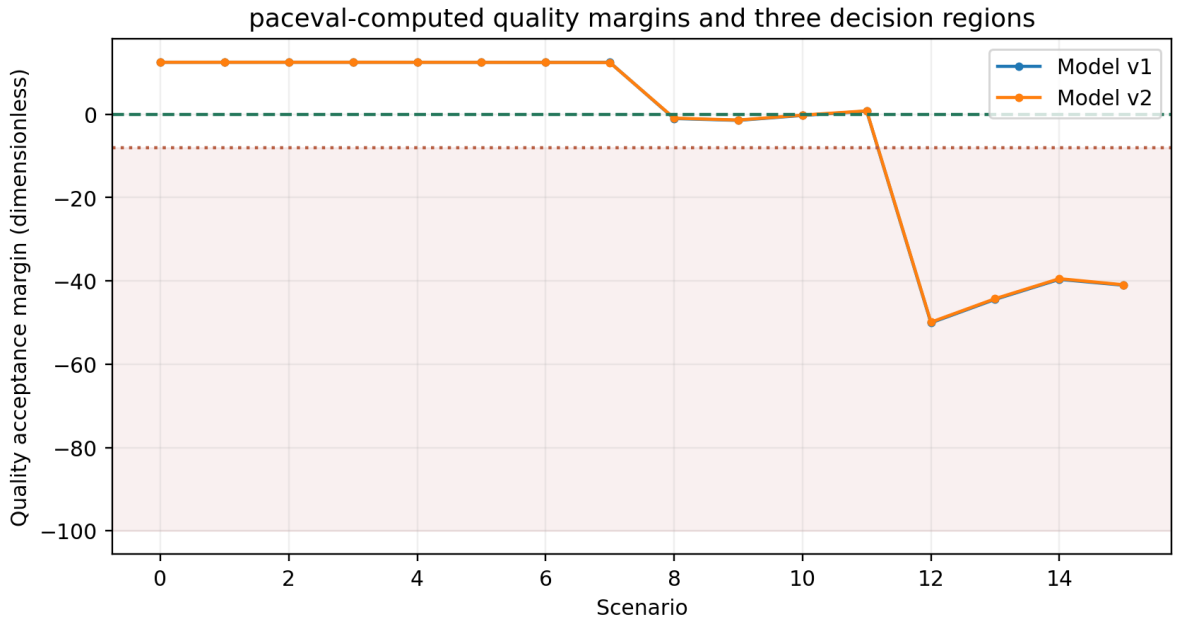


Figure 8. Signed acceptance margins evaluated in paceval. The host assigns the final research-only decision.

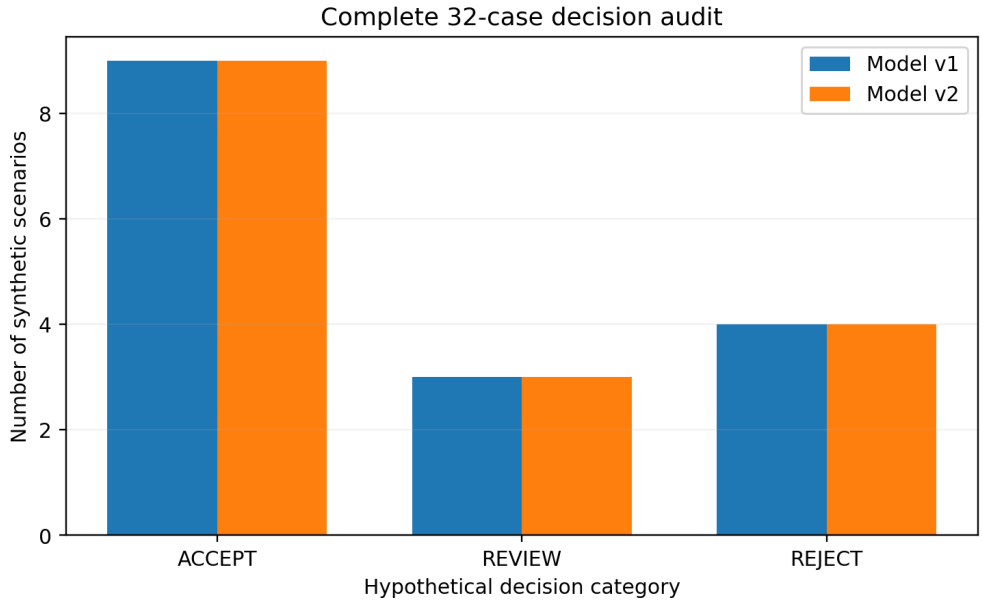


Figure 9. Synthetic quality outcomes from two independently preserved mathematical score versions.

10. Quality surface evaluated by paceval

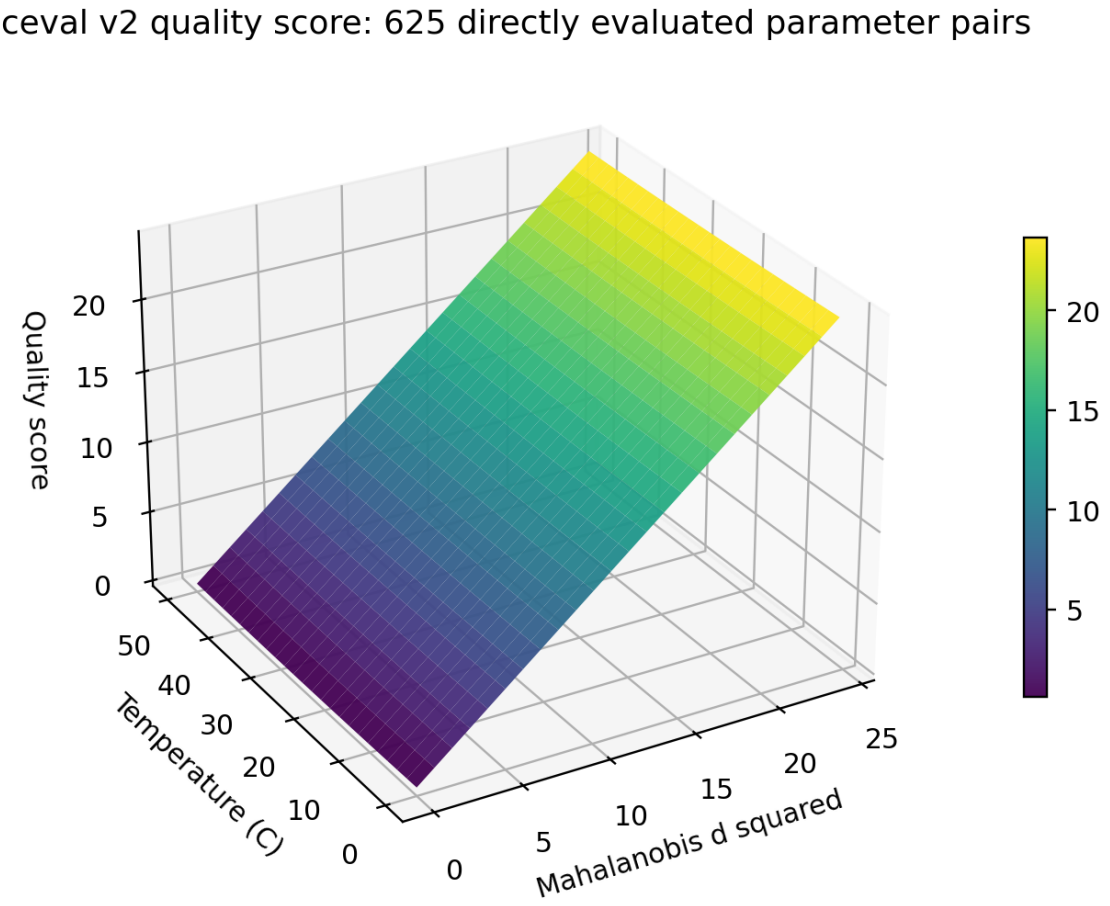


Figure 10. Genuine 625-point paceval score surface varying the innovation statistic and operating temperature, model v2.

A complete sweep of 25 × 25 points was submitted to the original paceval multi-graph batch API. The same inputs were evaluated across both score versions, two signed-margin expressions and two analytical derivative expressions. The surface uses the recorded 625-point result file; it is not an interpolated external reference calculation.

11. Industrial gate map and sensitivities

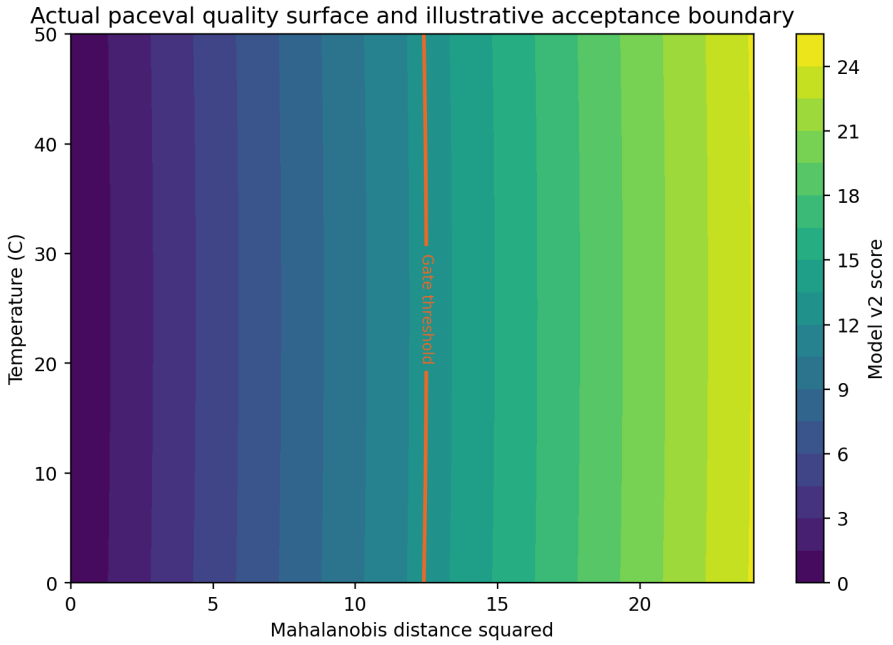


Figure 11. Executed model-v2 quality contours with the illustrative signed decision boundary.

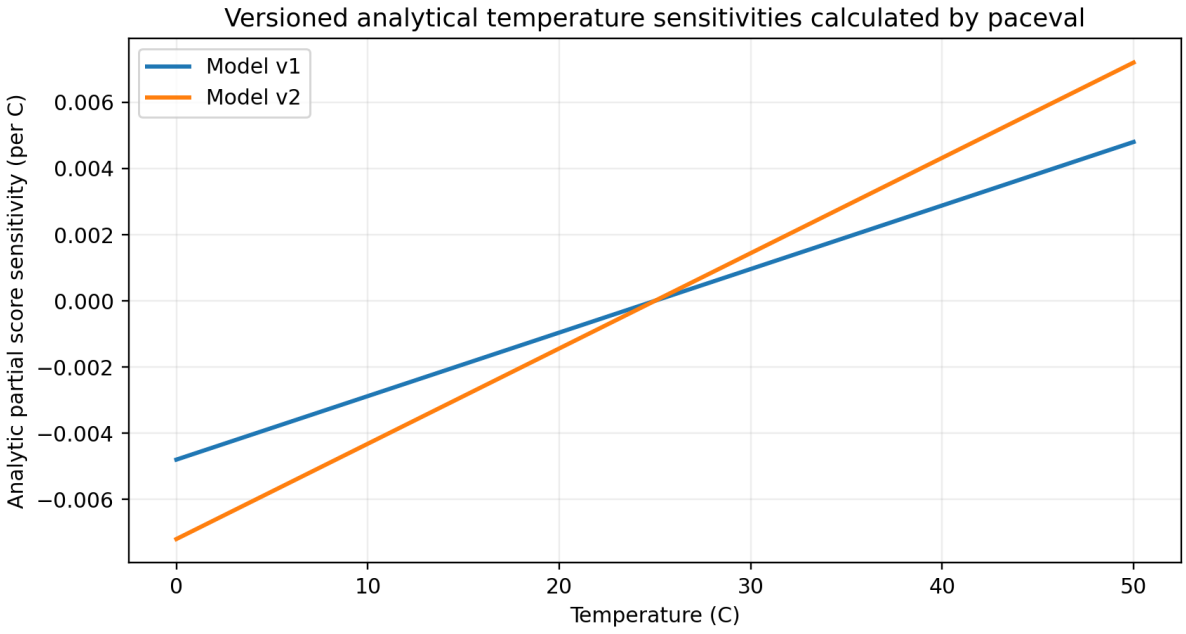


Figure 12. Analytic temperature derivatives from two separate actual paceval graphs.

12. Interpretation and research limitations

What is actually established

• Mathematical feasibility of fully populated dense SPD Cholesky, inversion and log-determinant through the original paceval API.

• Two-source covariance fusion including uncertainty and complete six-dimensional state estimates using reusable paceval graph kernels.

• Simultaneously available mathematical model versions, signed constraint margins and analytical score sensitivities.

• Independent complete-matrix, complete-fusion and selected high-precision checks, all passed.

What this study does not claim

No real industrial measurement data were used. The six-dimensional observation vectors, covariance coefficients and quality threshold are synthetic. The report therefore does not validate metrological traceability, safety performance, automated actuator control or deployment suitability for any specific machine or component.

The study does not investigate certified TINC interval bounds, arbitrary indefinite matrices, severe ill conditioning, production-grade hardware acceleration, or actual commercial productivity outcomes. The new scientific output is a verified execution and integration demonstration, not an assertion of superiority over specialized matrix libraries.

Next industrial research stage

A practical follow-up is to run an approved non-actuating shadow system with real multisensor metrology records, separately document physical units and sensor covariance calibration, reserve independent holdout equipment, validate measurement uncertainty, and adopt an approved version/rollback workflow for mathematical score rules. This is a proposed study, not a completed deployment.

13. Reproducibility and scientific references

Full reproduction from the package

Build: **cmake -S src -B build -DCMAKE_BUILD_TYPE=Release -DPACEVAL_RUNTIME=/absolute/path/to/libpaceval_linux_sharedLIB.so**; run **cmake --build build**, then **./build/matrix_sensor_fusion . --n 128** and **./build/matrix_sensor_fusion . --n 512**. Run **ctest --test-dir build --output-on-failure**.

Independent audit: **OPENBLAS_NUM_THREADS=1 python3 src/verify_results.py .**. Graph assignment: **python3 src/verify_inventory.py .**. Regenerate all scientific figures using **python3 src/make_figures.py .**.

Dated scientific and technical sources

[1] Netlib LAPACK, DPOTRF. Official documentation of blocked Cholesky factorization. https://www.netlib.org/lapack/explore-html/d2/d09/group__potrf_ga84e90859b02139934b166e579dd211d4.html

[2] Netlib LAPACK, DPOTRS. Official SPD equation-solving documentation. https://www.netlib.org/lapack/explore-html/d3/dc8/group__potrs_ga70a04d13ff2123745a26b1e236212cf7.html

[3] Peng et al. sqrtVINS: Robust and Ultrafast Square-Root Filter-based 3D Motion Tracking, 2025, arXiv 2510.10346. Preprint.

[4] Cao, Y. Multisensor Data Fusion-Driven Digital Twins in Computer Numerical Control Machining: A Review. Machines 13 (2025), 921. DOI: 10.3390/machines13100921.

[5] Frechette, Shao and Pease. Digital Twins Workshops Summary Report. NISTIR 8620 (July 2026). DOI: 10.6028/NIST.IR.8620.

[6] NIST, Digital Twins for Advanced Manufacturing program, 2024–2026. https://www.nist.gov/programs-projects/digital-twins-advanced-manufacturing

[7] Zhang et al. Hybrid digital twins for smart manufacturing: Architectures, fusion paradigm, and implementation challenges. Journal of Manufacturing Systems 85 (2026). DOI: 10.1016/j.jmsy.2025.12.029.

[8] Li et al. Towards adaptive closed-loop optimization: A review of sensor placement for digital twins in mechanical systems. Journal of Manufacturing Systems 88 (October 2026). DOI: 10.1016/j.jmsy.2026.07.027.

[9] paceval Mathematical Engine and public API. https://paceval.com/api/; https://github.com/paceval/paceval.

Research status as of 13 November 2026: executable original paceval research; synthetic datasets; no physical instrument certification or deployed industrial system.